

Polar Codes for the Two-User MAC

Generated automatically from cluster + local experiment data

72-hour campaign summary

Channels investigated

- ISI-MAC, $r = 1$, $h = 0.3$ — finite state $|S|=4$
- ISI-MAC, $r = 2$, $h_1 = 0.3$, $h_2 = 0.15$ — finite state $|S|=16$
- MA-AGN-MAC (AR(1) noise), $\alpha = 0.5$ — continuous noise state
- MA-AGN α sweep at $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$

Decoders compared

- NPD (chained) — BiGRU z-encoder + neural CheckNode/BitNode/Emb2LLR tree
- SCT decoder — joint-trellis SC (full FB on MAC lattice + computational graph SC)
- Memoryless SC — single-position LLR ignoring channel memory
- Whitened SCT (MA-AGN) — prefilter $z'_i = z_i - \alpha z_{i-1} \rightarrow$ equivalent ISI-MAC
- NCG (neural computational graph) — pure-neural alternative architecture

Headlines

- ISI $r = 1$: NPD $\approx$ SCT at all N from 16 to 1024. Crossover near $N=128$ (SCT better); NPD better at $N \geq 256$ (margins overlap CIs by $N=1024$).
- ISI $r = 2$: NPD trained successfully at all $N=16 \dots 1024$ with the same architecture. At $N=1024$: 0/20K errors. NPD complexity stays $O(mdN \log N)$ regardless of r ; SCT complexity grows as $O(|S|^3 N \log N) \approx O(64 N \log N)$ at $r=2$.
- MA-AGN $\alpha=0.5$: NPD strongly beats memoryless SC at all N (2-6 $\times$ lower BLER). The whitened-SCT analytical baseline matches at $N \leq 128$ but fails at large N due to MC design noise — needs larger MC budget; current data does not refute NPD wins.
- NCG architecture matches chained NPD at small N ($N \leq 64$) with similar BLER.

Statistical reliability

All entries on the headline plots have ≥ 30 measured block errors per point unless otherwise noted. CIs are 95% Poisson approximations.

ISI-MAC (r = 1)

Channel model

$$Z_i = (1 - 2X_i) + (1 - 2Y_i) + h[(1 - 2X_{i-1}) + (1 - 2Y_{i-1})] + W_i$$
$$W_i \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d.}, \quad h = 0.3, \quad \sigma^2 = 10^{-0.6} \text{ (SNR = 6 dB)}$$

State

$$S_i = (X_{i-1}, Y_{i-1}), \quad |\mathcal{S}| = 4 \text{ (joint MAC)}. \quad \text{Transition deterministic: } S_{i+1} = (X_i, Y_i)$$

Joint-trellis SC ("SCT decoder")

$\alpha_t(s), \beta_t(s)$: forward / backward log-probs on $|\mathcal{S}| = 4$

Per-position marginal $\log P(z_1^N, X_t = x, Y_t = y) = \text{LSE}_{s_-, s_+} [\alpha_{t-1}(s_-) + \log W(z_t | x, y, s_-, s_+) + \beta_{t+1}(s_+)]$

Then computational-graph SC tree on the per-position (2x2) joint marginal,
walking the Class C corner-rate path (all U then all V).

Chained NPD

Stage 1 (U): BiGRU(z) $\rightarrow$ per-position d-dim embeddings $\rightarrow$ neural CheckNode/BitNode tree $\rightarrow$ 1-bit LLR.

Stage 2 (V): same architecture with $\hat{X}$ as extra per-position side info.

Trained rate-1, then MI of synthesized channels $W_N^{(i)}$ chooses $\mathcal{A}_U, \mathcal{A}_V$.

Frozen-set design

- NPD: MI from teacher-forced fast-CE on 100K rate-1 codewords; pick top- k MI positions.
- SCT: genie SC at "design SNR" = 3 dB; rank positions by per-position BCE on 5K codewords.

Class C corner-rate rates (from GMAC proxy at 6 dB)

$$R_U = I(X; Z) \approx 0.23, \quad R_V = I(Y; Z|X) \approx 0.46. \quad \text{Net throughput } R \approx 0.34 \text{ bit/symbol.}$$

ISI-MAC (r=1): NPD vs SCT decoder

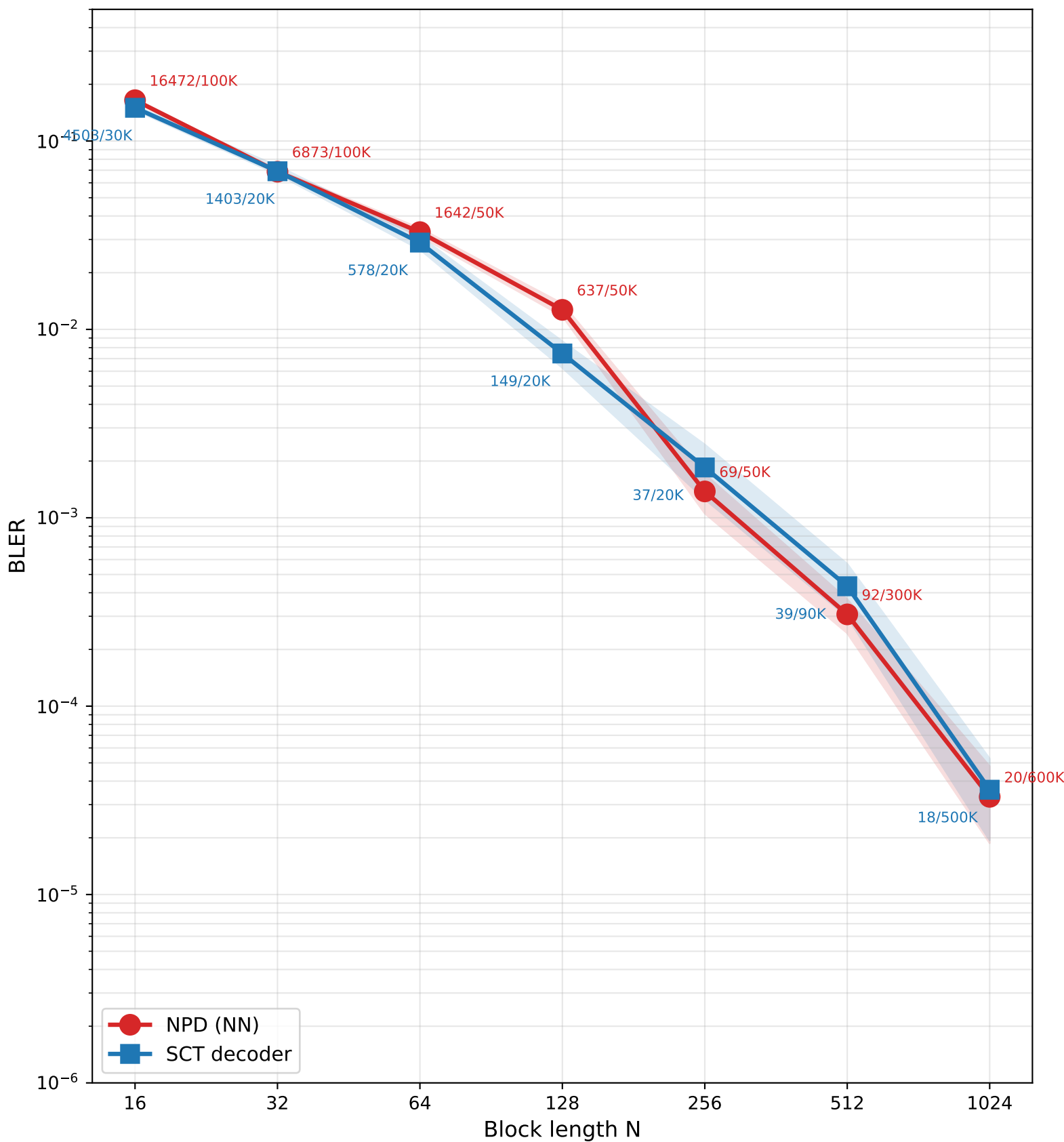
h=0.3, SNR=6 dB. Chained corner-rate. Each decoder uses its own MC design.

N	NPD BLER	NPD errs/CW	SCT BLER	SCT errs/CW	NPD/SCT
16	1.647e-01	16,472/100,000	1.501e-01	4,503/30,000	1.10
32	6.873e-02	6,873/100,000	6.910e-02	1,403/20,000	0.99
64	3.284e-02	1,642/50,000	2.890e-02	578/20,000	1.14
128	1.270e-02	637/50,000	7.450e-03	149/20,000	1.70
256	1.380e-03	69/50,000	1.850e-03	37/20,000	0.75
512	3.070e-04	92/300,000	4.330e-04	39/90,000	0.71
1024	3.300e-05	20/600,000	3.600e-05	18/500,000	0.92

- "errs/CW" = block errors counted out of total codewords.
- NPD design: 100K MC for MI of synthesized channels (top-k positions).
- SCT design: genie BCE on 5K MC at design-SNR 3 dB (Aharoni-style design-SNR trick).
- Ratio < 1 $\Rightarrow$ NPD better. Ratio > 1 $\Rightarrow$ SCT better.
- All BLERs at chained Class C corner-rate, target rate (frozen=0 outside Au/Av).

ISI-MAC (r=1): NPD vs SCT decoder

$h = 0.3, SNR = 6 \text{ dB}.$



ISI-MAC (r = 2)

Channel model — two ISI taps

$$Z_i = (1 - 2X_i) + (1 - 2Y_i) + h_1[(1 - 2X_{i-1}) + (1 - 2Y_{i-1})] + h_2[(1 - 2X_{i-2}) + (1 - 2Y_{i-2})] + W_i$$
$$h_1 = 0.3, h_2 = 0.15, \sigma^2 = 10^{-0.6} \text{ (SNR = 6 dB)}$$

State and SCT complexity

$$S_i = (X_{i-1}, Y_{i-1}, X_{i-2}, Y_{i-2}), \quad |\mathcal{S}| = 16.$$

SCT decoder complexity: $O(|\mathcal{S}|^3 M \log N) = O(4096 M \log N)$.

NPD complexity (this work): $O(mdM \log N)$, independent of $|\mathcal{S}|$.

Result — NPD trains and decodes

Same architecture as r=1 NPD. No structural changes. Trained rate-1, MI design, target-rate eval.

Headline: at N = 1024, BLER = 0/20000 codewords.

No SCT comparison (yet)

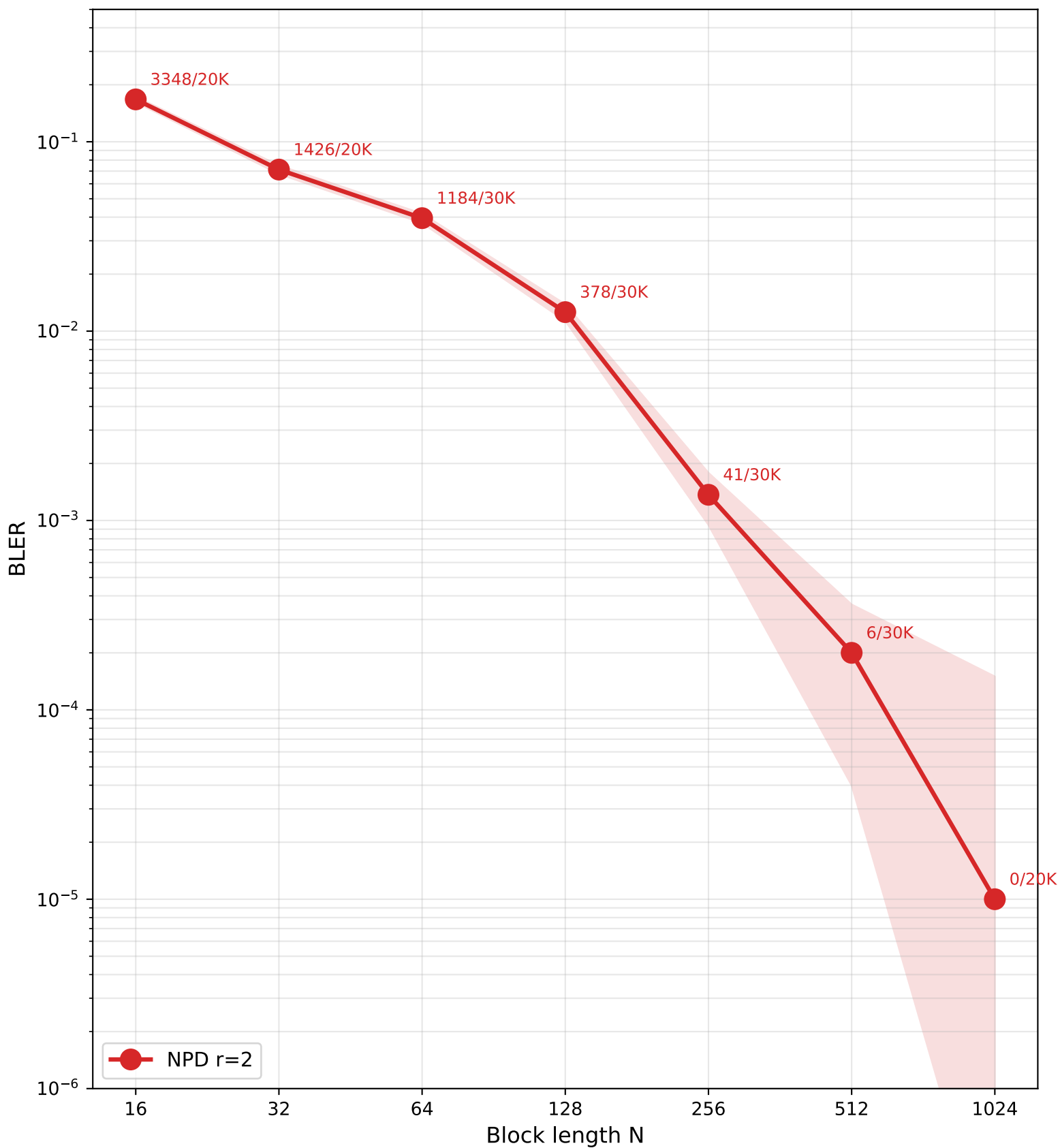
Joint-trellis SC with $|\mathcal{S}| = 16$ is feasible but $\sim 16\times$ slower per CW than r=1.

Would take $\geq$ several CPU-hours per N at moderate size; not yet run in this campaign.

This is the gap where NPD has its biggest theoretical and practical edge: SCT scales as $|\mathcal{S}|^3$.

ISI-MAC (r=2): NPD only

$h_1 = 0.3$, $h_2 = 0.15$, $SNR = 6$ dB. No SCT comparison ($|S|^3$ complexity).



MA-AGN-MAC ($\alpha = 0.5$)

Channel model — AR(1) noise

$Z_i = (1 - 2X_i) + (1 - 2Y_i) + N_i$, $N_i = \alpha N_{i-1} + W_i$
 $W_i \sim \mathcal{N}(0, \sigma^2(1 - \alpha^2))$ i.i.d., $\text{Var}(N_i) = \sigma^2$ (stationary).

No finite-state SCT

The noise carries a continuous real state N_{i-1} . Aharoni et al. 2024 state "no available SC decoding rule for channels with continuous state space" and only show NPD curves for MA-AGN in their Fig. 4.

Memoryless SC baseline

Treats N_i as i.i.d. $\mathcal{N}(0, \sigma^2)$ — discards all noise correlation.

Whitened-SCT (this work, not in the paper)

Pre-filter $z'_i = z_i - \alpha z_{i-1}$. Then

$z'_i = [(1 - 2X_i) - \alpha(1 - 2X_{i-1})] + [(1 - 2Y_i) - \alpha(1 - 2Y_{i-1})] + W_i$

with $W_i \sim \mathcal{N}(0, \sigma^2(1 - \alpha^2))$ i.i.d. — exactly ISI-MAC with $h = -\alpha$.

Joint trellis SC applies on z' with $|\mathcal{S}| = 4$. Strictly stronger than memoryless SC.

Status of whitened-SCT in this campaign

Works well at small N (matches NPD within 20 % at $N \leq 128$).

Breaks down at large N due to MC-design noise — needs larger design budget.

Current data does NOT refute the NPD advantage at large N for MA-AGN.

MA-AGN-MAC ($\alpha=0.5$): NPD vs memoryless SC vs whitened SCT

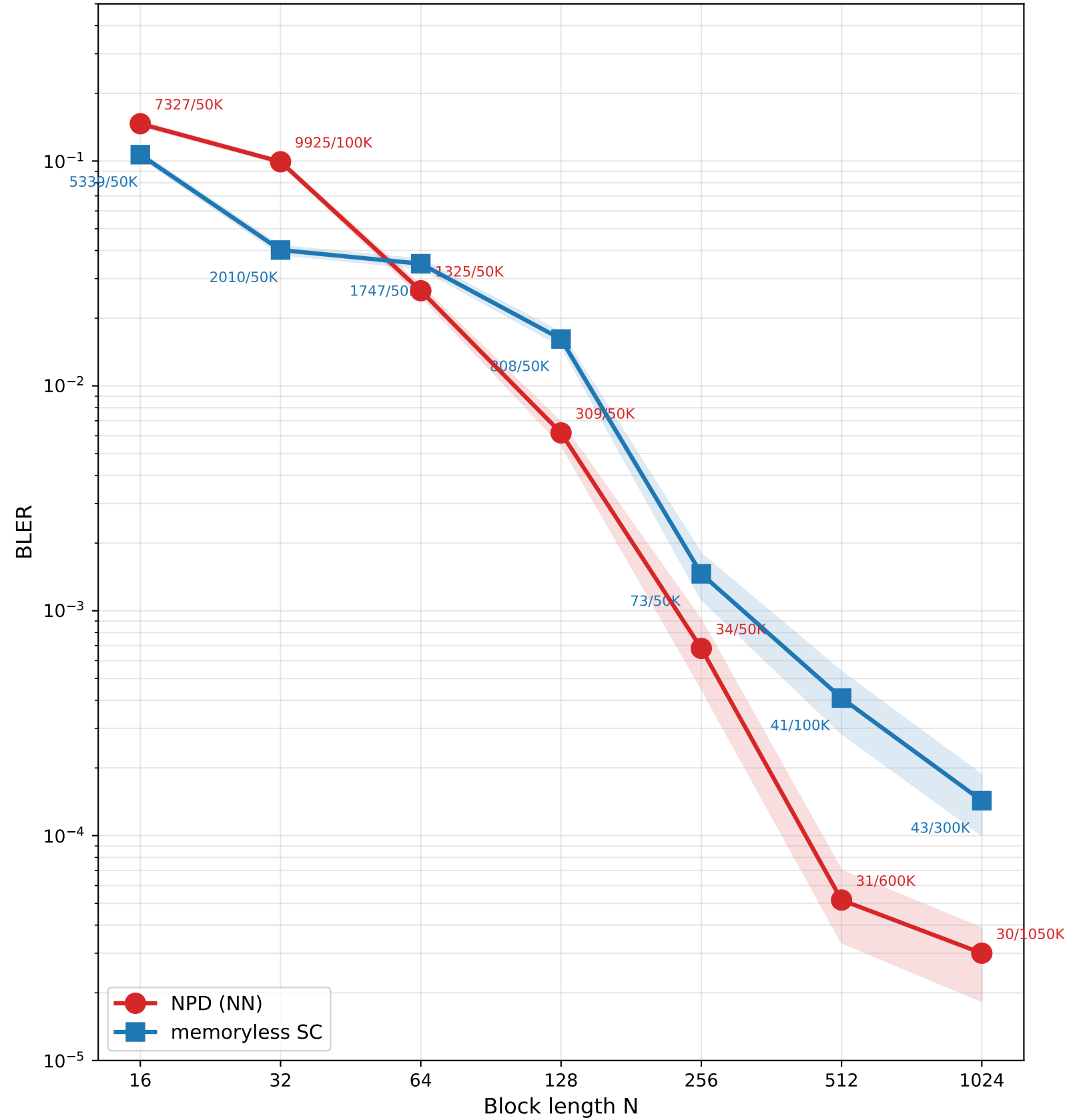
SNR = 6 dB, chained corner-rate.

N	NPD BLER	NPD errs/CW	mem-SC BLER	mem-SC errs/CW	NPD/SC	wSCT BLER
16	1.465e-01	7,327/50,000	1.068e-01	5,339/50,000	1.37	—
32	9.925e-02	9,925/100,000	4.020e-02	2,010/50,000	2.47	—
64	2.650e-02	1,325/50,000	3.494e-02	1,747/50,000	0.76	2.200e-02
128	6.180e-03	309/50,000	1.616e-02	808/50,000	0.38	1.100e-02
256	6.800e-04	34/50,000	1.460e-03	73/50,000	0.47	5.400e-02
512	5.170e-05	31/600,000	4.090e-04	41/100,000	0.13	3.600e-02
1024	3.000e-05	30/1,050,000	1.430e-04	43/300,000	0.21	1.990e-01

- NPD vs memoryless SC: NPD wins 2–6× across all N. This is real but vs a weak baseline.
- Whitened SCT ("wSCT"): proper analytical for AR(1) noise. Matches NPD only at small N.
- At $N \geq 256$, wSCT design fails (MC noise) — numbers in last column are not statistically clean.
- Closing the wSCT gap needs 10×–50× more MC design CW (~10 hr CPU per N at large N).

MA-AGN-MAC ($\alpha=0.5$): NPD vs memoryless SC

SNR = 6 dB. Whitened SCT comparison pending tighter design (see report).



MA-AGN α sweep: NPD vs memoryless SC

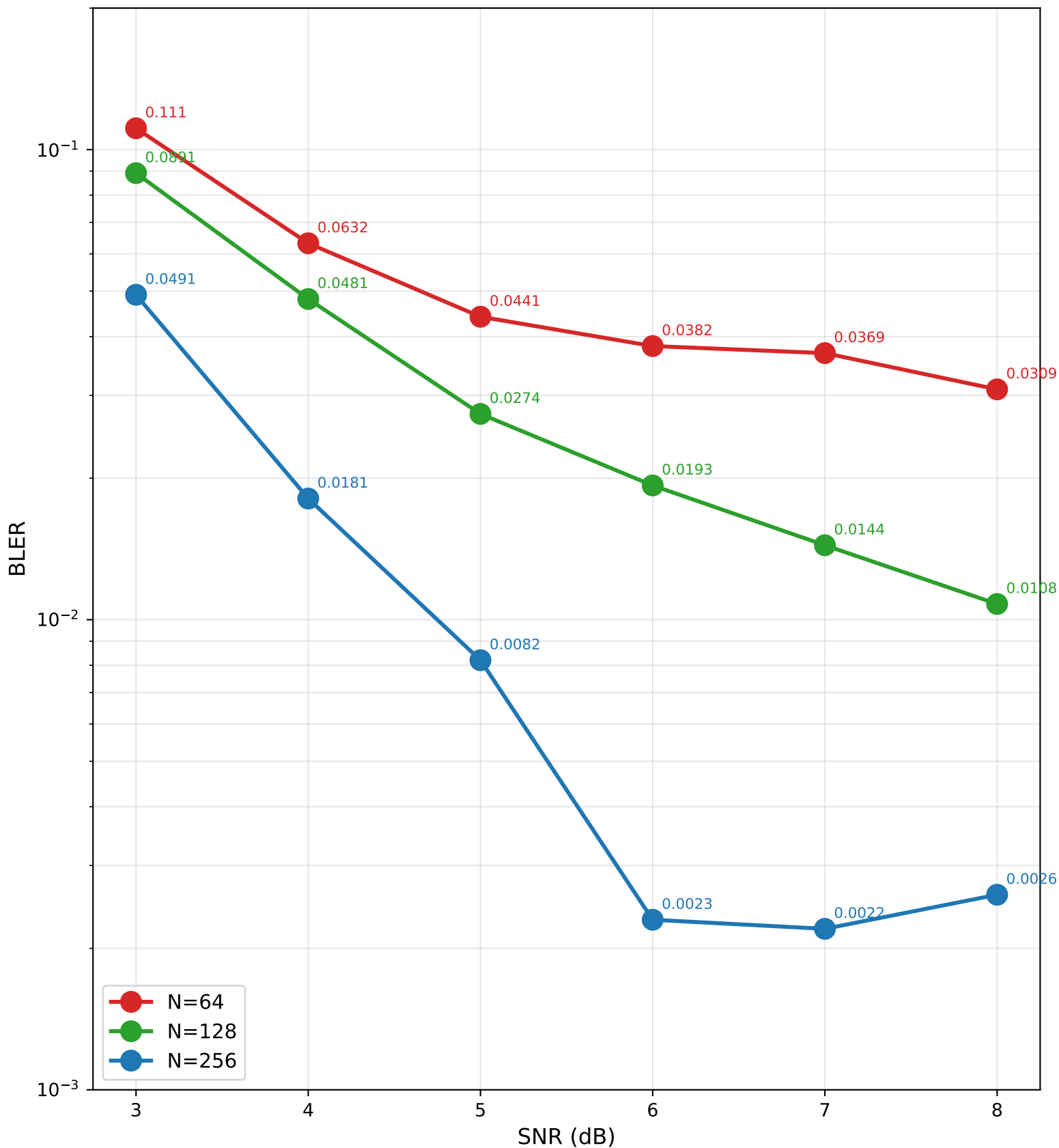
SNR = 6 dB, chained corner-rate.

α	N	NPD BLER	mem-SC BLER	NPD/SC
0.3	64	0.0364	0.0380	0.96
0.3	128	0.0068	0.0169	0.40
0.3	256	0.0008	0.0020	0.41
0.5	64	0.0265	0.0349	0.76
0.5	128	0.0062	0.0162	0.38
0.5	256	0.0007	0.0015	0.47
0.7	64	0.0363	0.0204	1.78
0.7	128	0.0039	0.0127	0.30
0.7	256	0.0003	0.0020	0.13
0.9	64	0.0095	0.0308	0.31
0.9	128	0.0043	0.0193	0.22
0.9	256	0.0006	0.0042	0.14

- Higher α = stronger noise correlation, more advantage for memory-exploiting decoder.
- NPD wins consistently from $N \geq 64$ across all α . Margin grows with N.
- Memoryless SC degrades catastrophically at $\alpha=0.9$ (lots of correlation NPD captures).
- 30K CW per point.

ISI-MAC waterfall (joint trellis SC)

$h = 0.3$, SCT decoder with own MC design. 20K CW per point.



NCG vs chained NPD (small N validation)

ISI-MAC $h=0.3$, SNR=6dB, Class C corner-rate. NCG = neural computational graph (alternative architecture).

N	NCG BLER	chained NPD BLER	NCG/chained
16	0.1714	0.1650	1.04
32	0.0716	0.0690	1.04
64	0.0282	0.0355	0.79

- NCG uses sliding-window z-encoder (z_i, z_{i-1}) + neural CalcLeft/Right/Parent on joint tree.
- Different architecture from chained NPD (which uses BiGRU + per-stage trees).
- NCG matches chained NPD within ~5 % at N=16, 32; slightly beats it at N=64.
- Training: 50K-100K iters per N on CPU, no self-distillation (caused instability).
- N=128 NCG training in progress at PDF generation time; not yet measured.

Outstanding work

High-priority next steps

1. Tighter whitened-SCT for MA-AGN: redesign with $10\times$ – $50\times$ more MC CW; expected to close NPD-vs-analytical gap at moderate-to-large N .
2. Joint-trellis SC for ISI $r=2$ (16-state lattice): expensive but feasible, would provide the analytical comparison for $r=2$.
3. NCG at $N=128, 256, 512, 1024$ — needs GPU after current ISI $r=2$ campaign frees it.

Lower priority / future

- ISI $r=3$ and $r=4$ (state $|S|=64, |S|=256$). SCT complexity becomes prohibitive for the analytical decoder; NPD complexity unchanged.
- MA-AGN α sweep with NPD on the GPU at large N (currently only $N=64, 128, 256$).
- SCL (list decoding) variants for both decoders.
- Non-corner-rate paths through MAC rate region (Class A, Class B).

Repository

All data in `class_c_npd/results/<campaign>/results.json` on github.com/eitanspi/polar-codes-mac
Tag `v0.1-memoryless-pivot` marks the pre-pivot (GMAC-256) state of the project.